

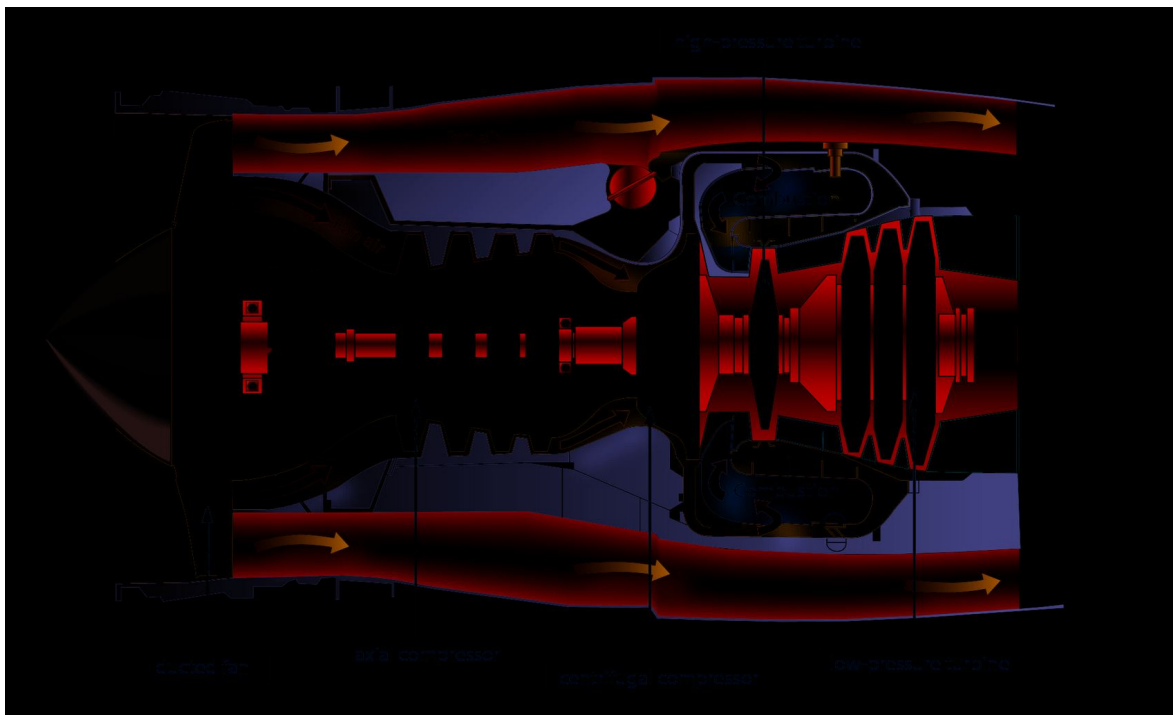


Figure 16-2. *Turbofan engine.*

Air drawn into the engine for the gas generator is further compressed and constitutes the core airflow. While a turbojet engine uses the entire gas generator's output to produce thrust in the form of a high-velocity exhaust gas jet, the lower velocity and cooler bypass air produces some of the thrust produced by a turbofan engine.

The turbofan engine design increases the thrust of the jet engine, particularly at lower speeds and altitudes. Although less efficient at higher altitudes, the turbofan engine increases acceleration, decreases the takeoff roll, improves initial climb performance, and often has the effect of decreasing fuel consumption.

Operating the Jet Engine

In a jet engine, the amount of fuel injected into the combustion chamber controls thrust. Because most engine control functions are automatic, the power controls on most turbojet-powered and turbofan-powered airplanes consist of just one thrust lever for each engine. The thrust lever links to a fuel control and/or electronic engine computer that meters fuel flow based on revolutions per minute (rpm), internal temperatures, ambient conditions, and other factors.

Typically in jet airplanes, there are flight deck indications for the rotation speed of each major engine section. Each engine section rotates at many thousands of rpm. For ease of interpretation, the indications read as percent of rpm rather than actual rpm. Depending on the make and model, there are usually indications for fuel flow, as well as for gas temperatures and pressures. The associated engine indications have different names according to their location.

As in any gas turbine engine, exceeding temperature or rpm limits, even for a few seconds, may result in serious damage to turbine blades and other components. The pilot should monitor the temperature of turbine gases and rotation speeds as needed. Modern aircraft are designed to prevent exceedances and alert the pilot of an impending or actual exceedance. Older designs rely more on the pilot to prevent any exceedances.

Setting Power

When setting power, the pilot normally uses pressure or rpm indications to set maximum allowable thrust. However, the forward movement of the thrust levers should be stopped for any limitation (e.g., pressure, rpm, or temperature).

Thrust to Thrust Lever Relationship

In a jet engine, thrust output changes much more per increment of throttle movement at high engine speeds. If the power setting is already high, it normally takes a small amount of movement to change the power output. This is a significant difference for the pilot transitioning to jet-powered airplanes. In a situation where significantly more thrust is needed and the jet engine is at low rpm, inching the thrust lever forward will have little effect. In this situation, the pilot needs to make a smooth and significant thrust lever position change to increase the power.

Variation of Thrust with RPM

Jets operate most efficiently in the 85 percent to 100 percent range. At idle rpm of approximately 55 percent to 60 percent, they produce a relatively small amount of thrust. An increase in rpm from 90 to 100 percent may increase thrust by as much as the total available at 70 percent. [Figure 16-3]

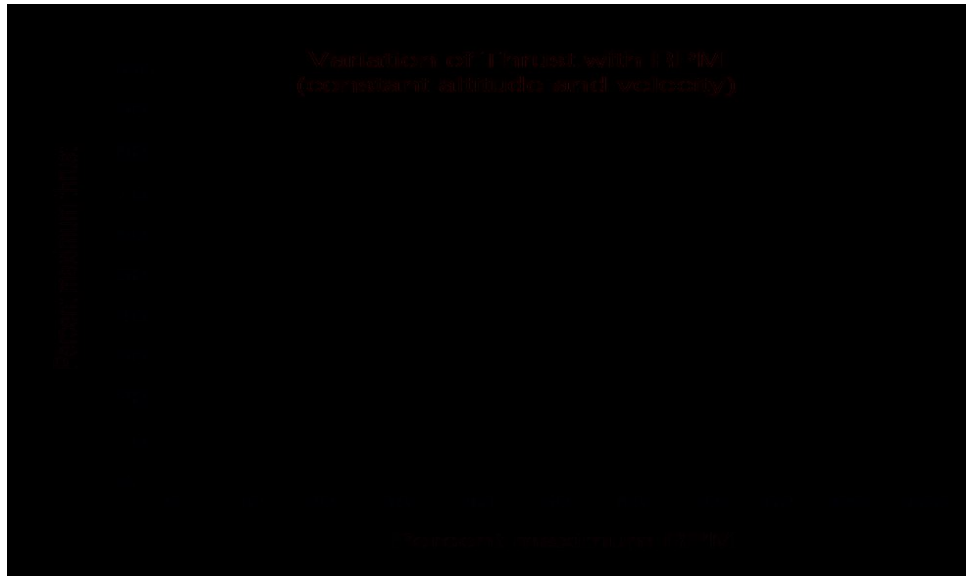


Figure 16-3. Variation of thrust with rpm.

Slow Acceleration of the Jet Engine

Acceleration of a piston engine from idle to full power is relatively rapid. The acceleration on different jet engines can vary considerably, but it is usually much slower. In some cases, the transition to full power could take up to 10 seconds. [Figure 16-4] Pilots should anticipate the need for adding power from low power settings.



Figure 16-4. Typical jet engine acceleration times.

Jet Engine Efficiency

The efficiency of the jet engine increases in the cold temperatures found at high altitudes. The fuel consumption of jet engines decreases as the outside air temperature decreases for constant engine rpm and true airspeed (TAS). Thus, by flying at a high altitude, the airplane operates with improved fuel economy and speed. At high altitudes, engines may be operating close to rpm or temperature limits, and excess thrust may not be available. Therefore, pilots should accomplish all maneuvering within the limits of available thrust, stability, and controllability.

Absence of Propeller Effects

The absence of a propeller affects the operation of jet-powered airplanes. Specific effects include the absence of lift from the propeller slipstream and the absence of propeller drag.

Absence of Propeller Slipstream

A propeller produces thrust by accelerating a large mass of air rearward. With wing-mounted engines, this air passes over a comparatively large percentage of the wing area. The total lift equals the sum of the lift generated by the wing area not in the wake of the propeller (as a result of airplane speed) and the lift generated by the wing area influenced by the propeller slipstream. By increasing or decreasing the speed of the slipstream air, it is possible to increase or decrease the total lift on the wing without changing airspeed. Since the jet airplane has no propellers, the transitioning pilot should note the following:

1. Lift is not increased instantly by adding power.
2. The stall speed is not decreased by adding power.

The lack of ability to produce instant lift in the jet, along with the slow acceleration of jet engines, necessitates a stabilized approach where landing configuration, constant airspeed, controlled rate of descent, and stable power settings are maintained until over the threshold of the runway. This allows for better engine response when making minor changes in the approach speed or rate of descent and improves go-around performance.

Absence of Propeller Drag

When the throttles are closed on a piston-powered airplane, the propellers create significant drag. Airspeed or altitude is immediately decreased. The effect of reducing power to idle on the jet engine, however, produces no such drag effect. In fact, at an idle power setting, the jet engine still produces forward thrust. While this can be an advantage in certain descent profiles, it is a handicap when it is necessary to lose speed quickly. The lack of propeller drag, along with the aerodynamically clean airframe of the jet, are new to most pilots, and slowing the airplane down is one of the initial problems encountered by pilots transitioning into jets. In level flight at idle power, it takes about 1 mile to lose 10 knots of airspeed.

Speed Margins

Maximum speeds in jet airplanes are expressed differently and always define the maximum operating speed of the airplane, which is comparable to the VNE of the piston airplane. These maximum speeds in a jet airplane are referred to as:

- V_{MO} —maximum operating speed expressed in terms of knots.
- M_{MO} —maximum operating speed expressed as a Mach number (the decimal ratio of true airspeed to the speed of sound).

Mach number is the ratio of true airspeed to the speed of sound. The speed of sound varies with temperature. At low/warm altitudes, the speed of sound is so high that an aircraft is limited by indicated airspeed. At high/cold altitudes, the speed of sound is lower so the aircraft is limited by Mach. To observe both limits V_{MO} and M_{MO} , the pilot of a jet airplane needs both an airspeed indicator and a Mach indicator. In most jet airplanes, these are combined into a single display for airspeed and Mach number, as appropriate.

It looks much like a conventional airspeed display with the addition of a "barber pole" that automatically moves so as to indicate the applicable speed limit at all times. [Figure 16-5]

A jet airplane can easily exceed its speed limitations. The handling qualities of a jet may change significantly at speeds higher than the maximum allowed.

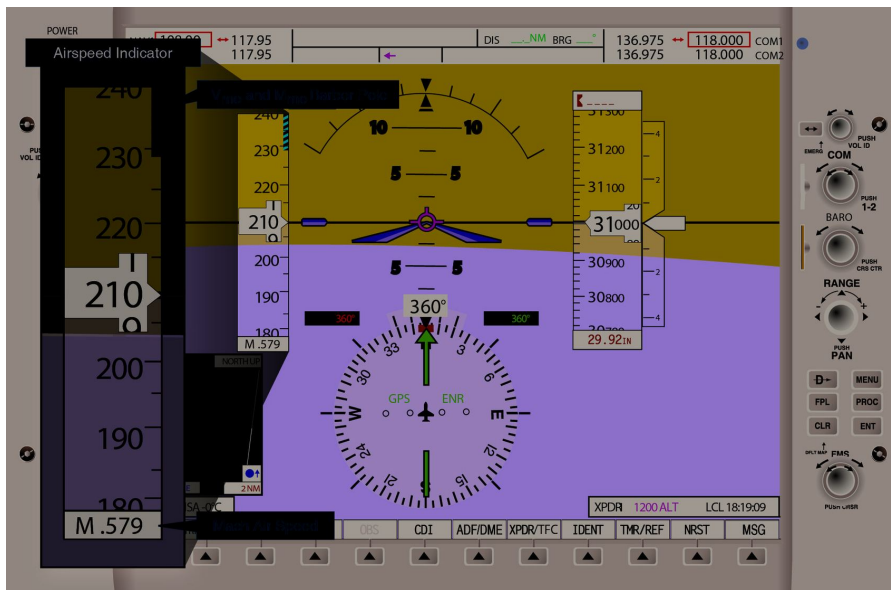


Figure 16-5. Jet airspeed indicator.

High-speed airplanes designed for subsonic flight are limited to some Mach number below the speed of sound. Shock waves (and the adverse effects associated with them) can occur when the airplane speed is substantially below Mach 1.0. The Mach number at which some portion of the airflow over the wing first equals Mach 1.0 is termed the critical Mach number (M_{CR}).

There is no particular problem associated with the acceleration of the airflow up to the critical Mach number, the point where Mach 1.0 airflow begins. However, a shock wave is formed at the point where the airflow suddenly returns to subsonic flow. This shock wave becomes more severe and moves aft on the wing as airflow velocity increases. Eventually, flow separation occurs behind the well-developed shock wave. [Figure 16-6]

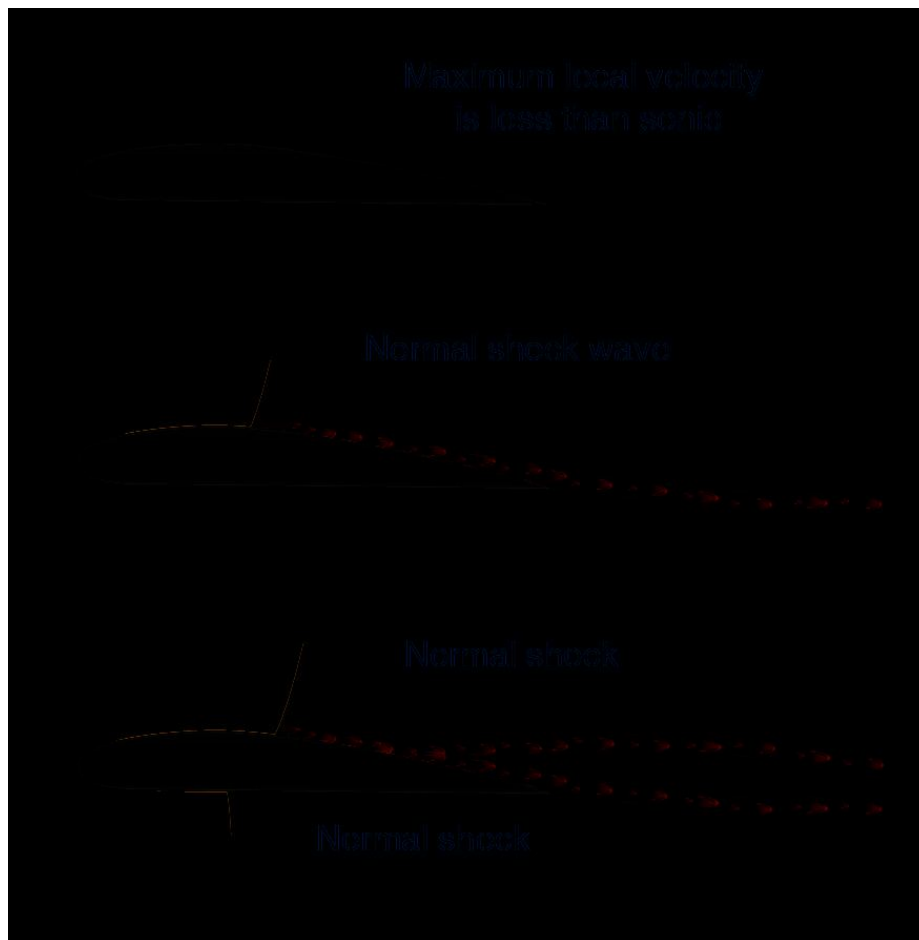


Figure 16-6. Transonic flow patterns.

If airplane speed progresses sufficiently beyond M_{MO} , the separation of air behind the shock wave may result in severe buffeting and possible loss of control or “upset.” Because of the accompanying changes to the center of lift, the airplane may exhibit pitch change tendencies.

With increased speed and the aft movement of the shock wave, the wing’s center of pressure moves aft causing the start of a nose-down tendency or “tuck.” Mach tuck develops gradually, and the condition should not be allowed to progress to where there is no longer enough elevator authority to prevent entry into a steep, sometimes unrecoverable, dive. An alert pilot should respond to excessive airspeed, buffeting, or warning devices before the onset of extreme nose-down forces.

Due to the critical aspects of high-altitude/high-Mach flight, most jet airplanes capable of operating in the Mach ranges use some form of automated Mach tuck compensation. If the system becomes inoperative, the airplane is typically limited to a reduced maximum Mach number.

Mach Buffet

Mach buffet arises when airflow separates on the upper surface of a wing behind a shock wave. All other things being equal, shock wave strength increases as the local airflow speed ahead of the shock wave increases. Mach buffet is a function of the speed of the airflow over the wing—not necessarily the forward speed of the airplane, and the shock wave strength, rather than a stall, creates the airflow separation.

Mach buffet may result from two different conditions in cruise. At high-speed cruise, a shock wave that becomes too strong as the airflow speeds up over the upper surface causes a buffet. At low-speed cruise, the flow has a greater turn to make to follow the wing’s upper surface. The air speeds up to do that and may exceed Mach 1 over the upper surface.

The shock wave position is different between the two situations. At high speed and a lower AOA, the shock wave tends to move aft. So when the flow separates behind the shock, that separated flow acts over a small range of the chord. In some cases, the separated flow acting on a small surface area may produce a little buzz. At low-speed cruise, the true airspeed is still high, but the shock wave does not move as far aft as it does in high-speed cruise. The separated flow behind the shock wave acts over a larger portion of the chord, which leads to a more significant effect on aircraft control.

The altitude at which an airplane flying at M_{MO} would experience buffeting with any increase in AOA determines the absolute or aerodynamic ceiling. This is the altitude where:

- If an airplane flew any faster, it would exceed M_{MO} leading to high-speed Mach buffet.
- If an airplane flew any slower, it would require an angle of attack leading to low-speed Mach buffet.

This region of the airplane’s flight envelope is known as “coffin corner.” Conceivably, a buffet could be the first indication of an issue at altitude, and pilots should understand the cause of any buffet in order to respond appropriately.

An increase in load factor (G factor) will raise the low-end buffet speed. For example, a jet airplane flying at 51,000 feet altitude at 1.0 G and a speed of 0.73 Mach that experiences a 1.4 G load, may encounter low-speed buffet. Consequently, a maximum cruising flight altitude and speed should be selected, which will allow sufficient margin for maneuvering and turbulence. The pilot should know the manufacturer’s recommended turbulence penetration speed for the particular make and model airplane. This speed normally gives the greatest margin between the high-speed and low-speed buffets.

Low-Speed Flight

The jet airplane wing, designed primarily for high-speed flight, has relatively poor low-speed characteristics. As opposed to the normal piston-powered airplane, the jet wing has less area relative to the airplane’s weight, a lower aspect ratio (long chord/short span), and thin airfoil shape—all of which amount to the need for speed to generate enough lift. The swept wing is additionally penalized at low speeds because its effective lift is proportional to airflow speed that is perpendicular to the leading edge.

In a typical piston-engine airplane, V_{MD} (minimum drag) in the clean configuration is normally at a speed of about $1.3 V_S$. [Figure 16-7] Flight below V_{MD} in a piston-engine airplane is well identified and predictable. In contrast, in a jet airplane, flight in the area of V_{MD} (typically $1.5 - 1.6 V_S$) does not normally produce any noticeable changes in flying qualities other than a lack of speed stability—a condition where a decrease in speed leads to an increase in drag, which leads to a further decrease in speed, which creates the potential for a speed divergence. A pilot who is not aware of a developing speed divergence may find a serious sink rate developing at a constant power setting, while pitch attitude appears to be normal. The fact that lack of speed stability may lead to a sinking flightpath, is one of the most important aspects of jet-airplane flying.